Think Where It Matters: Role-Specific Thinking Mode Allocation in Multi-Agent LLM Debate*

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Abstract

Reasoning-capable large language models and multi-agent debate systems have each advanced AI problem-solving, yet their intersection remains unexplored. We present a systematic ablation study of role-specific thinking mode allocation in a three-agent (Proposer, Critic, Judge) debate framework. Evaluating all eight thinking-mode configurations across three model families (Gemini 2.5 Flash, Claude Sonnet 4, OpenAI o4-mini) and three benchmarks (GSM8K, MATH-500, GPQA-Diamond), we demonstrate that the all-ON configuration is suboptimal in most settings. Selective thinking allocation not only reduces cost but can improve accuracy: on MATH-500, OpenAI's Judge-only thinking achieves 81.0% versus 78.0% for all-ON, while costing only 1.09x the baseline. Our results reveal that optimal allocation depends on model family, task difficulty, and agent role, with verification roles (Critic, Judge) generally benefiting more than the generation role (Proposer). These findings suggest that reasoning capacity should be treated as a selectively allocable resource, particularly in operational settings where cost, reliability, and degraded operating conditions matter. We provide practical guidelines for cost-efficient and cyber-resilience-aware deployment of thinking-enabled multi-agent systems.

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1 Introduction

The emergence of reasoning-capable large language models has marked a significant shift in AI problem-solving. Models such as DeepSeek-R1 [1], OpenAI o1 and o3 [2], and Gemini 2.5 Flash [3] incorporate explicit chain-of-thought reasoning through thinking modes, allocating additional compute at inference time to produce intermediate reasoning traces. These modes have demonstrated remarkable improvements on challenging benchmarks in mathematics, science, and coding.

Concurrently, multi-agent systems (MAS) built on LLMs have gained traction as a means to improve output reliability. The debate paradigm, in which multiple agents argue for and against proposed solutions before a judge renders a verdict, has proven particularly effective [4, 5]. By exposing reasoning to adversarial critique, debate systems catch errors that single-agent approaches miss.

These converging trends necessitate a critical examination of role-specific reasoning allocation; specifically, whether the collective intelligence of a multi-agent debate system truly requires every participant to operate in thinking mode. While activating deep reasoning across all agents dramatically increases computational expenditure—often by a factor of three to five—the marginal utility of such uniform allocation remains empirically unverified. If a proposer requires deep reasoning to navigate a vast solution space, does a critic tasked with error detection necessitate the same depth? Similarly, does a judge's ability to synthesize perspectives depend more on its own internal reasoning or on the quality of the reasoning traces provided by the agents it evaluates? From a cyber resilience perspective, this allocation problem is especially important: operational AI systems may need to preserve reliable reasoning under resource constraints, degraded conditions, or adversarial pressure. Selectively assigning thinking capability to verification-oriented roles can provide a practical mechanism for maintaining decision quality without incurring the cost of full-system reasoning.

Despite rapid proliferation of both reasoning models and multi-agent frameworks, this question has received surprisingly little attention. Existing work on heterogeneous MAS [6, 7] focuses on mixing different model backbones rather than manipulating reasoning depth within a single model. Which specific roles within a MAS benefit most from thinking mode activation—and whether targeted allocation outperforms uniform activation—remains unexplored.

In this paper, we present a systematic ablation study of role-specific thinking allocation in a three-agent debate framework. We evaluate all eight possible thinking-mode configurations ($2^3=8$) across three model families and three benchmarks. Our contributions are: (1) the systematic ablation of role-specific thinking allocation in multi-agent debate; (2) cross-model validation across three model families, revealing model-dependent and difficulty-dependent optimal strategies; and (3) practical guidelines for cost-efficient deployment, showing that targeted allocation can substantially reduce costs while preserving or improving accuracy.

2 Related Work

2.1 Multi-Agent Debate

Du et al. [4] demonstrated that multi-LLM debate significantly improves factuality and reasoning. Subsequent work explored various topologies: Liang et al. [5] introduced divergent thinking structures, while Chan et al. [8] examined multi-agent debate for improved LLM-based evaluation. CONSENSAGENT [9] addressed sycophancy in multi-agent debate, and DiMo [10] explored diverse reasoning strategies using prompt-based approaches.

2.2 Thinking Models and Reasoning

The thinking model paradigm, popularized by OpenAI's o1 [2], demonstrates that allocating additional inference-time compute to explicit reasoning chains yields substantial gains. This has been adopted across providers: DeepSeek-R1 [1] introduced open-weight reasoning models, Gemini 2.5 Flash [3] offers configurable thinking budgets, and Claude models [11] support extended thinking with controllable token budgets. A key shared feature is the ability to toggle thinking within the same checkpoint, enabling controlled experiments.

2.3 Heterogeneous Multi-Agent Systems

X-MAS [6] investigates heterogeneous LLM backbone assignment. MasRouter [7] dynamically routes queries to different model tiers. MALBO [12] frames MAS design as Pareto optimization over cost and performance. While these approaches manipulate the model backbone, our work manipulates reasoning depth within a single model—a complementary and previously unexplored dimension.

2.4 Thinking Modes in Multi-Agent Settings

While thinking models and multi-agent debate have each been studied extensively, their intersection remains largely unexplored. Existing multi-agent frameworks uniformly apply the same reasoning mode to all participants, treating thinking as a system-wide toggle. Our work introduces a finer-grained dimension: which specific roles benefit from thinking, and whether targeted allocation is more efficient than the all-or-nothing approach.

3 Method

3.1 Debate Architecture

We employ a three-agent, two-round debate structure. The Proposer receives the problem and generates an initial solution with step-by-step reasoning. The Critic examines the Proposer's answer, identifies errors, and suggests corrections; if correct, it provides independent verification. In round two, the Proposer revises its answer incorporating feedback, and the Critic re-evaluates. Finally, the Judge synthesizes the complete exchange and renders a final answer. This yields five API calls per debate (Proposer and Critic each twice, Judge once).

3.2 Thinking Mode Configurations

Each agent possesses a binary thinking mode state where ON denotes that the thinking mode is enabled and OFF denotes it is disabled. Consequently, the configuration space for the entire three-agent system yields $2^3 = 8$ possible configurations, labeled C0 through C7 (Table 1). For instance, configuration C5 indicates that the Proposer and Judge utilize thinking mode for deep reasoning, while the Critic operates in standard generation mode. Within this operational space, C0 (all OFF) serves as the system baseline, and C7 (all ON) represents the upper bound of computational cost. These intermediate configurations (C1–C6) allow us to isolate the specific contribution of each role's reasoning process to the collective performance and identify the most cost-efficient allocation strategies.

Table 1. Thinking Mode Configurations

Config	Proposer	Critic	Judge	Label
C0	OFF	OFF	OFF	All OFF
C1	ON	OFF	OFF	P only
C2	OFF	ON	OFF	Cr only
C3	OFF	OFF	ON	J only
C4	ON	ON	OFF	P+Cr
C5	ON	OFF	ON	P+J
C6	OFF	ON	ON	Cr+J
C7	ON	ON	ON	All ON

3.3 Models

We evaluate three model families, each offering native thinking mode control within a single checkpoint. Gemini 2.5 Flash supports a configurable thinking_budget parameter (0 = OFF, 8192 = ON). Claude Sonnet 4 offers an extended thinking parameter that can be enabled with a budget of 8,192 tokens or omitted. OpenAI o4-mini provides a reasoning_effort parameter ("high" = ON, "low" = reduced); notably, o4-mini cannot fully disable reasoning. Using three models from distinct providers strengthens generalizability of observed patterns.

3.4 Benchmarks

We select three benchmarks spanning a wide difficulty spectrum. GSM8K (Easy) consists of 200 randomly sampled (seed=42) grade-school math problems from the test set [13]. MATH-500 (Hard) comprises 200 sampled competition-level mathematics problems [14, 15]. GPQA-Diamond (Expert) uses all 198 graduate-level multiple-choice science questions, where domain experts achieve ~65% accuracy [16]. This spectrum allows examination of how thinking allocation interacts with task complexity.

3.5 Evaluation Metrics

Our primary metric is accuracy via exact match. For GSM8K, we extract numerical values; for MATH-500, we normalize LaTeX expressions; for GPQA-Diamond, we match selected options (A/B/C/D).

We additionally track total token consumption as a cost proxy, including input tokens, output tokens, and thinking tokens when enabled. To ensure comparability across configurations, cost is normalized by the all-OFF (C0) configuration within each model. We compute cost-efficiency as accuracy gain per unit cost increase over the all-OFF baseline.

3.6 Experimental Controls

We hold constant the problem set, ordering, prompt text, debate structure, answer extraction logic, and thinking token budget across all configurations and models. Temperature is set to 0 where the API permits. Where temperature cannot be fixed (Claude with thinking enabled), we apply the same default across all configurations for that model.

4 Results

4.1 RQ1: Which Roles Benefit Most from Thinking?

Tables 2–4 report accuracy across all eight configurations for each model–benchmark pair.

Table 2. Accuracy (%) on GSM8K (Easy)

Config	Gemini	Claude	OpenAI
C0 (All OFF)	96.0	96.5	95.5
C1 (P)	96.5	96.0	96.0
C2 (Cr)	96.0	95.5	95.5
C3 (J)	96.5	96.0	96.5
C4 (P+Cr)	97.0	96.5	96.0
C5 (P+J)	97.0	95.5	96.5
C6 (Cr+J)	96.5	96.0	96.0
C7 (All ON)	97.5	95.5	95.5

Table 3. Accuracy (%) on MATH-500 (Hard)

Config	Gemini	Claude	OpenAI
C0 (All OFF)	90.0	83.5	69.5
C1 (P)	90.5	68.5	74.0
C2 (Cr)	90.0	88.5	73.5
C3 (J)	90.5	84.0	81.0
C4 (P+Cr)	91.5	85.0	76.0
C5 (P+J)	91.0	72.0	78.5
C6 (Cr+J)	90.5	87.5	78.5
C7 (All ON)	91.0	86.0	78.0

Table 4. Accuracy (%) on GPQA-Diamond (Expert)

Config	Gemini	Claude	OpenAI
C0 (All OFF)	78.7	74.2	73.2
C1 (P)	80.3	76.7	75.2
C2 (Cr)	80.3	78.2	76.7
C3 (J)	79.2	77.7	76.2
C4 (P+Cr)	80.8	79.2	79.7
C5 (P+J)	79.7	76.2	77.2
C6 (Cr+J)	79.7	80.3	78.2
C7 (All ON)	80.3	82.3	76.7

On GSM8K, thinking allocation has minimal impact across all models, with accuracy ranges of at most 3.0 percentage points (pp). No configuration significantly outperforms the all-OFF baseline, confirming that thinking mode on easy tasks often represents pure overhead and that reasoning is unnecessary for tasks already solvable by base model competence.

Conversely, the performance dynamics shift significantly on the MATH-500 benchmark. For OpenAI o4-mini, Judge-only thinking mode (C3) achieves 81.0%, outperforming the all-ON configuration (C7, 78.0%) by 3.0pp—a striking case where less thinking yields better performance. This counter-intuitive result suggests that for certain architectures, excess reasoning tokens may introduce stochastic noise rather than logical clarity. For Claude, Critic-only thinking (C2) achieves 88.5% (+5.0pp over C0), while Proposer-only thinking (C1) causes a severe 15.0pp degradation to 68.5%. For Gemini, the Proposer+Critic combination (C4) achieves the highest accuracy of 91.5%.

On GPQA-Diamond, the most challenging benchmark, Claude shows the widest range (8.1pp), with the all-ON configuration (C7) achieving the best result at 82.3%. OpenAI shows a 6.5pp range, with the Proposer+Critic (C4) peaking at 79.7%. Gemini remains remarkably stable (2.6pp range).

4.2 RQ2: Does Optimal Strategy Vary with Difficulty?

A clear interaction between task difficulty and optimal thinking allocation emerges. On easy tasks (GSM8K), thinking provides no consistent benefit; thus, C0 represents the most cost-efficient choice. On hard tasks (MATH-500), the mean accuracy gap between the worst and best configurations is 9.5pp (C1: 76.0% to C6: 85.5%). Critically, C7 (all-ON, 85.0%) is outperformed by partial configurations, notably C6 (Cr+J, 85.5%). On expert tasks (GPQA-Diamond), while more thinking generally helps, the cross-model mean of C4 (P+Cr, 80.1%) marginally outperforms that of C7 (79.7%) while incurring only two-thirds of the thinking cost.

4.3 RQ3: Cross-Model Comparison

Optimal thinking allocation is strongly model-dependent. Table 5 summarizes the optimal configuration by model and difficulty.

Table 5. Optimal Configuration by Model and Difficulty

Model	Easy (GSM8K)	Hard (MATH-500)	Expert (GPQA)
Gemini 2.5 Flash	C7 (97.5%)	C4 (91.5%)	C4 (80.8%)
Claude Sonnet 4	C0/C4 (96.5%)	C2 (88.5%)	C7 (82.3%)
OpenAI o4-mini	C3 (96.5%)	C3 (81.0%)	C4 (79.7%)

Gemini 2.5 Flash exhibits strong baseline performance (90.0% on MATH-500, 78.7% on GPQA without thinking), limiting the headroom for improvement. Its best configuration (C4) on MATH-500 yields only +1.5pp over C0, suggesting that Gemini's base model already incorporates sufficient implicit reasoning.

Claude Sonnet 4 shows the most dramatic sensitivity. On MATH-500, Proposer-only thinking (C1) causes a catastrophic 15.0pp drop. We hypothesize that this degradation reflects a mismatch between reasoning verbosity and downstream evaluative capacity. A thinking mode-enabled Proposer produces substantially longer reasoning traces, which a non-thinking Critic may fail to effectively process or critique, thereby weakening the adversarial feedback loop.

OpenAI o4-mini shows a distinctive Judge-centric pattern. On MATH-500, Judge-only thinking (C3) achieves 81.0%—outperforming both C0 (+11.5pp) and C7 (+3.0pp). This is the single most cost-efficient configuration in our study: thinking in only one of five agent invocations yields the highest accuracy.

4.4 All-ON Suboptimality

A central finding is that C7 (all-ON) is suboptimal in the majority of settings. In five of six non-trivial model–benchmark pairs (excluding GSM8K), at least one partial configuration outperforms C7 (Table 6). The sole exception is Claude on GPQA-Diamond.

Table 6. Cases Where Partial Thinking Outperforms C7

Model	Benchmark	Best Config	Acc (%)	C7 Acc (%)
Gemini	MATH-500	C4 (P+Cr)	91.5	91.0
Gemini	GPQA	C4 (P+Cr)	80.8	80.3
Claude	MATH-500	C2 (Cr)	88.5	86.0
OpenAI	MATH-500	C3 (J)	81.0	78.0
OpenAI	GPQA	C4 (P+Cr)	79.7	76.7

4.5 Cost-Efficiency Analysis

The practical significance of role-specific thinking is best understood through a cost-accuracy trade-off analysis.

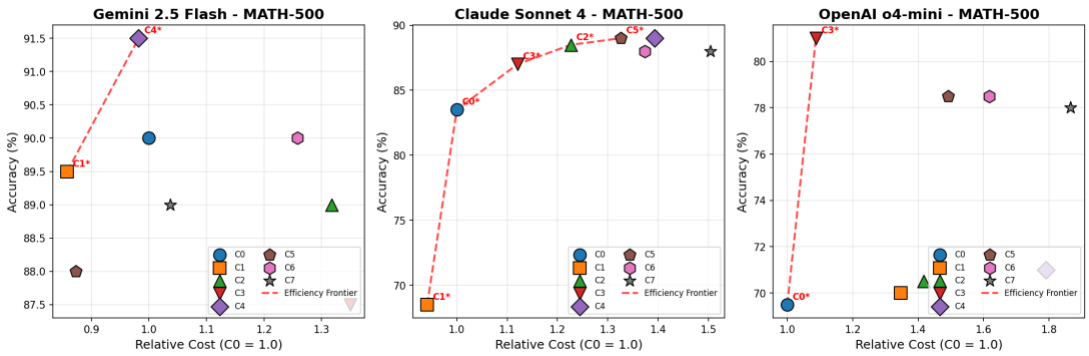


Figure 1. Cost-accuracy trade-off by model on MATH-500.

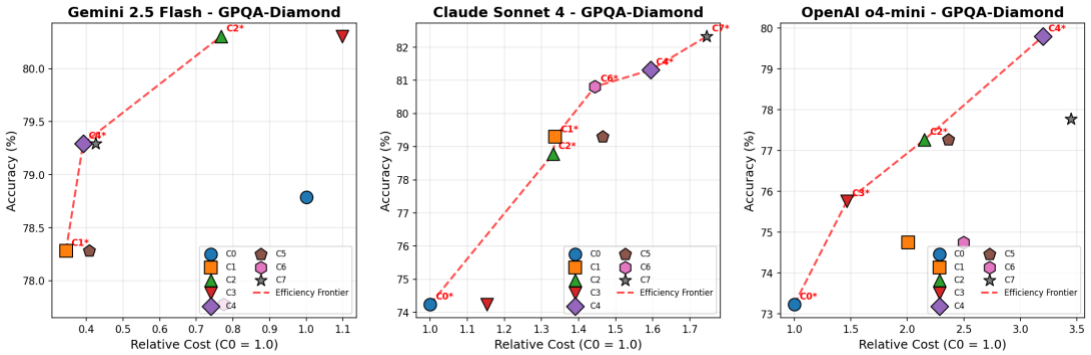


Figure 2. Cost-accuracy trade-off by model on GPQA-Diamond.

Figure 1 illustrates this relationship for the MATH-500 benchmark. The visualization highlights the "efficiency frontier"—the set of configurations that deliver the highest accuracy for a given cost. For Gemini, C4 (Proposer+Critic) emerges as the dominant strategy, achieving peak accuracy (91.5%) with virtually no cost increase (0.98x) over the baseline. Claude follows a similar trajectory, where C4 provides the best performance-to-cost balance.

A standout finding is observed in OpenAI o4-mini: the C3 (Judge-only) configuration yields a dramatic 11.5 percentage point improvement at a marginal 1.09x cost. This suggests that for certain architectures, a single, high-quality reasoning step at the synthesis stage is far more effective than distributing reasoning across all agents.

Figure 2 shifts the focus to the expert-level GPQA-Diamond benchmark, where the efficiency patterns change significantly. Unlike the math tasks, expert-level scientific reasoning appears to demand a higher “thinking density.” For instance, Claude reaches its peak performance (82.3%) only in the C7 (All-ON) configuration. This indicates that as task complexity scales from competition math to graduate-level science, the system requires more comprehensive reasoning from all participants to resolve deep ambiguities, even if it entails higher costs.

As summarized in Table 7, these results demonstrate that “thinking more” is not always better. By identifying model-specific sweet spots—like OpenAI’s Judge-centric efficiency or Claude’s need for All-ON in expert tasks—practitioners can bypass the diminishing returns of a uniform All-ON approach.

Table 7. Cost-Efficiency on MATH-500 (Mean)

Config	Mean Acc (%)	Mean Cost (x)	Acc/Cost
C0 (All OFF)	81.0	1.00x	baseline
C3 (J only)	85.2	1.09x	46.7
C6 (Cr+J)	85.5	1.67x	6.7
C4 (P+Cr)	84.2	1.65x	4.9
C7 (All ON)	85.0	2.75x	2.3

5 Discussion

5.1 Mechanisms of Suboptimality

We hypothesize that the underperformance of the 'All-ON' (C7) configuration stems from Reasoning Anchoring. When a Proposer generates an excessively verbose, thinking-augmented response, it may inadvertently 'anchor' the subsequent Critic. A Critic operating under the influence of such a dense reasoning chain may struggle to maintain the adversarial independence required to spot subtle logical fallacies, essentially being 'led' by the Proposer's logic.

To provide empirical evidence for this hypothesis, we analyzed experimental logs where C0 (All-OFF) succeeded but C1 (Proposer-only thinking) failed. In Gemini's GPQA results, we identified 15 such cases. For instance, in problem #18 involving organic chemistry (Michael reaction), C0 correctly predicted answer 'C', while C1 incorrectly predicted 'A'. The C1 Proposer generated an elaborate analysis of regioselectivity, but made an error in interpreting steric hindrance effects. The Judge acknowledged that "the Proposer initially made an error," yet ultimately followed the Proposer's flawed reasoning chain. Similarly, in problem #29 (photochemistry), the C1 Proposer's excessive Stokes shift analysis led to answer 'C' instead of the correct 'A'. These cases demonstrate that verbose extended thinking can anchor subsequent agents to flawed reasoning paths.

Furthermore, we identify Reasoning Redundancy as a primary cost-driver with diminishing returns. In many-agent systems, if every participant independently reconstructs the same underlying logic, the Judge is presented with a redundant information landscape that complicates synthesis rather than clarifying it. This explains why the Judge-only (C3) configuration in models like o4-mini often yields superior results: a single, high-fidelity reasoning step at the final decision point is often sufficient to resolve ambiguities without the noise of redundant intermediate traces.

Quantitative analysis supports this observation. On GSM8K, enabling thinking for all agents (C7) increased costs by 229% for Gemini (from $0.0005 to $0.0017 per problem) while improving accuracy by only 1.5 percentage points. For Claude, the 68% cost increase actually resulted in a 1.0 percentage point accuracy decrease (96.5% to 95.5%), and OpenAI showed zero accuracy gain despite 158% higher costs. Interestingly, on GPQA, the optimal thinking allocation varies significantly across models. Claude benefits from All-ON (C7), achieving 82.3% accuracy, suggesting that its extended thinking effectively handles complex scientific reasoning. In contrast, Gemini achieves its best performance (80.3%) with C2 (Critic-only thinking), and OpenAI peaks at 79.8% with C4 (Proposer+Critic). This divergence suggests that for Gemini and OpenAI, the Judge benefits from receiving cleaner, less verbose inputs from the debate, rather than processing additional thinking tokens.

5.2 Asymmetric Value of Thinking Mode by Role

Verification roles (Critic and Judge) generally benefit more from thinking than the generation role (Proposer). This aligns with complexity-theoretic arguments: verifying a solution is computationally easier than generating one, but verification quality depends critically on analysis depth. A thinking Critic more effectively identifies subtle errors, while a thinking Judge more carefully weighs conflicting arguments. Conversely, the Proposer's task may not require the same explicit reasoning depth, particularly when the model's implicit capabilities are strong.

5.3 Cost-Accuracy Trade-off Landscape

Figures 3 and 4 provide a unified visualization of the cost-accuracy trade-offs across all three models on MATH-500 and GPQA-Diamond benchmarks. This visualization reveals distinct model characteristics: Gemini clusters in the upper region regardless of configuration, indicating robust

baseline performance and lower marginal gains from thinking mode. Claude shows moderate sensitivity to thinking allocation, while OpenAI demonstrates the widest accuracy range (69.5% to 81.0%), suggesting high sensitivity to strategic thinking placement.

The combined analysis supports our key finding: optimal thinking allocation is model-dependent. Practitioners should consider both the target model's characteristics and the task difficulty when selecting configurations.

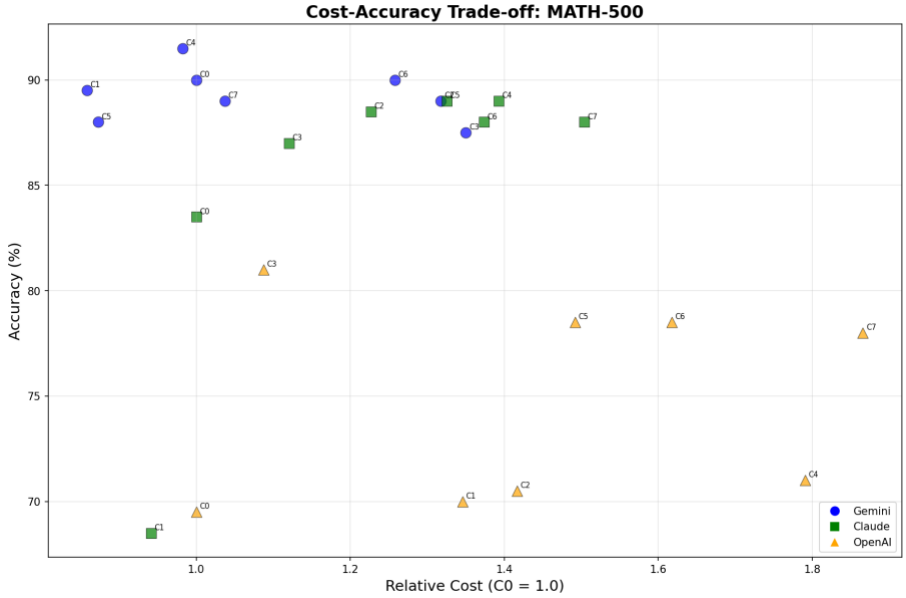


Figure 3. Unified cost-accuracy trade-off across all models on MATH-500.

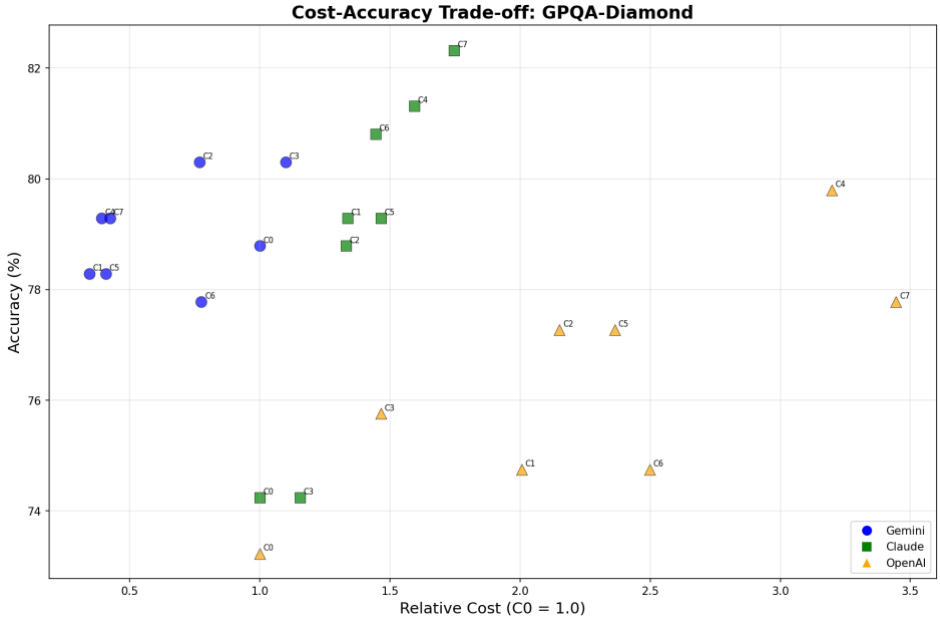


Figure 4. Unified cost-accuracy trade-off across all models on GPQA-Diamond.

5.4 Practical Guidelines

Based on our empirical results, the most effective deployment strategies require a nuanced approach rather than a one-size-fits-all solution. For easy tasks where baseline accuracy exceeds 95%, activating thinking mode is generally counterproductive, as it increases latency and cost without meaningful performance gains; in these cases, thinking should be disabled entirely.

When tackling hard tasks, the optimal strategy diverges by model. For OpenAI o4-mini, the C3 (Judge-only) configuration offers exceptional efficiency, delivering peak performance with minimal additional expenditure. In contrast, for Gemini and Claude, the C4 (Proposer+Critic) combination is recommended to best balance reasoning depth with robust verification. As tasks scale to expert-level complexity (e.g., GPQA-Diamond), the C4 (Proposer+Critic) configuration remains a strong cross-model performer, providing the necessary reasoning density to resolve graduate-level scientific ambiguities.

Ultimately, practitioners should move away from the assumption that the All-ON (C7) configuration is inherently superior. Instead, they should actively evaluate partial thinking allocations to identify the Pareto-optimal point that best fits their specific model, budget, and task complexity.

6 Limitations and Future Work

Our study has several limitations. First, all experiments are single-run without repeated trials, so variance estimates are unavailable. While sample sizes (198–200 problems per cell) provide reasonable power for detecting large effects, small inter-configuration differences should be interpreted cautiously. Future work should include repeated trials with statistical significance testing to validate the robustness of observed differences. Second, OpenAI o4-mini cannot fully disable reasoning; its "low" setting represents reduced rather than absent thinking, making C0 for o4-mini not strictly comparable to the fully-off configurations of Gemini and Claude. Third, our debate structure is fixed at three agents and two rounds; different topologies and round counts may interact differently with thinking allocation. Fourth, the thinking budget is fixed at 8,192 tokens; varying it continuously could reveal more nuanced trade-offs. Fifth, we evaluate only reasoning-intensive benchmarks; applicability to creative writing, coding, or open-ended QA remains unexplored.

Future work should address these limitations through repeated trials with variance estimation, continuous thinking budget exploration, diverse debate topologies, and broader task types. Automated methods for selecting optimal thinking allocations based on difficulty estimation or model confidence represent a promising practical direction.

7 Conclusion

We have presented the systematic study of role-specific thinking mode allocation in multi-agent LLM debate. Through exhaustive evaluation of eight thinking configurations across three model families and three benchmarks, we demonstrate that selective thinking allocation is not merely a cost-saving heuristic but a genuine performance optimization. Our key finding—that the all-ON configuration is suboptimal in most settings—challenges the assumption that more reasoning is always better in multi-agent systems. The optimal allocation depends on a complex interaction between model family, task difficulty, and role within the debate. For practitioners, our results offer immediately actionable guidance: disable thinking on easy tasks, start with Judge-only or Critic+Judge configurations on hard tasks, and always validate against partial configurations before committing to the all-thinking mode default.

Beyond cost efficiency, these findings have implications for cyber-resilient AI system design. In real-world cyber environments, AI systems may operate under constrained budgets, degraded infrastructure, adversarial inputs, or incident-response pressure. Our results suggest that resilience should not be equated with maximizing reasoning everywhere. Instead, cyber-resilient multi-agent systems may benefit from preserving deeper reasoning in roles responsible for verification and final adjudication, such as the Critic and Judge, while avoiding unnecessary reasoning redundancy across all agents. By thinking where it matters, multi-agent systems can achieve comparable or superior accuracy at dramatically reduced cost.

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